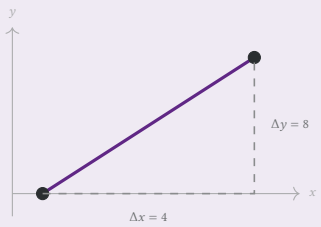
QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 3

RULE · Slope (qiyalik)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Term bridge: traditional textbooks call this k (and $y = kx + b$); the SAT calls it **slope** and m (and $y = mx + b$) – same thing.

Example: through $(3, 2)$ and $(7, 10)$:
$m = \frac{10 - 2}{7 - 3} = \frac{8}{4} = 2$



In $y = mx + b$: m is the slope, b is the y-intercept. Don't mix them up: in $y = 5x - 3$, slope is 5, y-intercept is -3.

WARNING · Don't invert the formula

Always **rise** (y) **over run** (x): $m = \frac{y_2 - y_1}{x_2 - x_1}$ – **never** $\frac{x_2 - x_1}{y_2 - y_1}$.

RULE · Writing a line's equation

Point-slope form: $y - y_1 = m(x - x_1)$

Slope 4, **point** $(1, 3)$:
$y - 3 = 4(x - 1) \Rightarrow y - 3 = 4x - 4 \Rightarrow y = 4x - 1$

Two points $(2, 1)$ **and** $(4, 7)$: find m first: $m = \frac{7 - 1}{4 - 2} = 3$; then use $(2, 1)$:
$y - 1 = 3(x - 2) \Rightarrow y - 1 = 3x - 6 \Rightarrow y = 3x - 5$

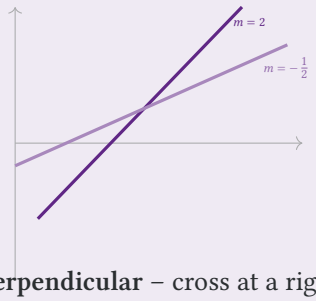
Slope-intercept form ($y = mx + b$) is best for graphing and reading the slope; point-slope form is best for starting from a point and a slope.

RULES SHEET – 2/2

RULE · Parallel & perpendicular lines

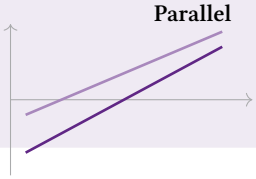
$m_1 = m_2$ (parallel) $\qquad$ $m_1 \times m_2 = -1$ (perpendicular)

Perpendicular to $y = 2x - 5$, **through** $(4, 1)$:
Negative reciprocal of slope 2: $-\frac{1}{2}$.
$y - 1 = -\frac{1}{2}(x - 4) \Rightarrow y = -\frac{1}{2}x + 3$

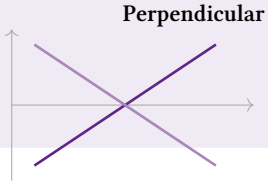


Parallel lines – same slope, different y-intercept (never cross). **Perpendicular** – cross at a right angle.

Parallel



Perpendicular



WARNING · Flip AND negate

A perpendicular slope is both flipped and sign-reversed: $\frac{2}{3} \to -\frac{3}{2}$ – don't just flip and forget the sign.

RULE · Real-world meaning of slope & intercept

In a word problem: **slope** = rate of change (per unit), y-**intercept** = starting value (at $x = 0$).

Example: $C(x) = 2000x + 5000$ (taxi fare, x = km). $C(10) = 2000(10) + 5000 = 25000$ som. (2000 = price per km; 5000 = starting fee.)

WARNING: read the question fully – is it asking for the slope itself, the value at a given x, or x itself? Each needs a different move.

The sign matters too: a negative slope means decrease (e.g. water draining from a tank), a positive slope means increase.